Supplementary material

Wang & Shih, "Identification of a weak muscle group tolerates more joint-angle error than kinematic reliability conventions allow: a predictive-simulation study"

Face-validity, provenance, sensitivity and interval checks, with the feature, classifier and reference specifications. Every number comes from a named script, listed in S13, run on the frozen 66-condition dataset except where an earlier rebuild is named (S4).

Contents.

S1. Face validity of the imposed deviations

Each imposed deviation was compared with the consequence the clinical literature reports; knee angle is negative in flexion in this model. Each check is evaluated inside its own gait phase: the whole-cycle ankle maximum is mid-stance tibial progression, not swing clearance, and the whole-cycle knee minimum is peak swing flexion, not the loading-response yield. Phases were segmented per limb from the vertical ground reaction force (> 0.05 BW), *early* and *late* being the first and last 40 % of stance, with each check averaged over strides.

On the unimpaired base model: stance 59.4 % of the cycle (healthy 60–62 %); peak swing knee flexion 58.0° averaged over strides, 58.3° as the single minimum over the whole window (the value quoted in S9); peak swing dorsiflexion 5.2°; peak push-off plantarflexion −10.0°; terminal-stance knee +2.0°; peak swing hip flexion 45.5° against a published ~25–35° (S9).

S1.1 Results by gait phase

Imposed	Expected (phase)	vs non-paretic	vs own unimpaired	mean Δ vs non-paretic
Hip flexors	reduced peak hip flexion (swing)	14/14	14/14	−9.8°
Plantarflexors	reduced peak plantarflexion (terminal stance)	15/15	13/15	+4.7°
Knee extensors	greater peak knee flexion (swing)	10/10	10/10	−5.2°
Knee extensors	reduced knee extension (terminal stance)	9/10	9/10	−4.0°
Dorsiflexors	reduced peak dorsiflexion (swing)	15/15	10/15	−12.5°

Knee flexors	genu recurvatum (terminal stance)	12/12	8/12	+4.7°
Knee flexors	swing knee flexion little changed	11/12	6/12	+1.5°

Denominators differ by row because accepted conditions are not evenly spread across severities.

Against each variant's own unimpaired gait:

Check (vs own unimpaired)	**20 %**	**40 %**	**60 %**
Hip flexors, peak swing hip flexion	5/5	5/5	4/4
Knee extensors, peak swing knee flexion	5/5	4/4	1/1
Knee extensors, terminal-stance extension	4/5	4/4	1/1
Plantarflexors, peak push-off plantarflexion	4/5	4/5	5/5
Dorsiflexors, peak swing dorsiflexion	2/5	3/5	5/5
Knee flexors, genu recurvatum	2/5	4/5	2/2
Knee flexors, swing flexion little changed	2/5	3/5	1/2

Every row that fails at 20 % improves by 60 % except the knee-flexor swing row (3 of 5 at 40 %, 1 of 2 at 60 %); of the three rows marked ǂ in Table 1, it is the one that is not dose-dependent.

Peak swing dorsiflexion, fifteen dorsiflexor conditions, degrees (negative = plantarflexed during swing):

	20 %	**40 %**	**60 %**
base	+7.2	−15.1	−16.1
s1	+6.0	+7.9	−16.6
s2	+9.6	+9.9	−12.5
s3	+4.9	−13.2	−17.0
s4	+3.1	+5.0	−13.5

At 60 % every variant has crossed into plantarflexion during swing: foot drop in the clinical sense. Below it the paretic ankle stays in dorsiflexion, and the 15/15 contralateral asymmetry comes from the sound side increasing its dorsiflexion, which is compensation rather than the imposed deficit. The threshold is not imposed by the analysis.

S1.2 Effect of the analysis window

Check	**whole cycle**	**correct phase**	**why**
Dorsiflexors, vs own unimpaired	5/15	10/15	whole-cycle ankle maximum is mid-stance

			tibial progression, not swing clearance
Plantarflexors, vs non-paretic	11/15	15/15	whole-cycle ankle minimum can be the small plantarflexion after initial contact
Knee extensors, loading response	9/10	4/10	the whole-cycle knee minimum is peak *swing* flexion, not the loading-response yield
Plantarflexors, ankle range of motion	6/15	withdrawn	range of motion is not the quantity Kesar 2009 measured

The knee-extensor row moved *against* our own expectation: the expected yield appears in only 4 of 10 conditions, and on average the knee is *less* flexed under load (mean +2.2°). Vasti can be weakened by 80 % or removed with normal kinematics still trackable (van der Krogt 2012 [2]), but that study prescribed kinematics and does not predict the direction of the residual. The consistent directions are the vasti's own events: eccentric restraint of early-swing knee flexion (10/10, mean 8.0° more flexion against own unimpaired gait) and knee extension in terminal stance (9/10, mean 4.2° less extension), the latter a documented flexed-knee pattern after stroke.

S1.3 Literature notes and caveats

The withdrawn ankle range-of-motion row (S1.2). Against each variant's own unimpaired solution, paretic ankle excursion *rises* in eleven of fifteen conditions (base variant 20.5° → 23.9°); the asymmetry exists only because the sound side rises further, to 29.5°.

Dorsiflexor and both knee-flexor rows hold against own unimpaired gait in two-thirds or fewer of conditions, so part of that asymmetry is sound-side adaptation. The knee-flexor rows also sit on divided literature: Okada 2024 found knee flexor strength lower in the clusters with the longest hyperextension durations, Cooper 2012 found the only significant association to be plantarflexor weakness in midstance ($p = 0.044$). The two are not head-to-head. Cooper compared patients with and without hyperextension, Okada duration clusters among patients who all had it, in twenty and thirty patients so the row is reported, not claimed as validation.

Reported peak push-off plantarflexion in healthy adults spans 8–20°, driven by walking speed (7.85° slow to 16.1° fast; Kwon 2015) and by foot model: under a multi-segment (Oxford Foot) model at normal speed it is −0.3° hindfoot-against-tibia but −12.7° forefoot-against-tibia (Sun 2018), a rigid foot, used here, lumping the two rotations into one ankle angle. Most stroke-versus-healthy comparisons are not speed-matched, so part of every published deficit is a speed confound. Produced by `face_validity_phase.py`; the whole-cycle values by `face_validity_all.py`.

S2. Optimisation depth and between-variant variation

Analysed optimisation depth differed by variant, with a median analysed generation of 293 for base (range 17–1469), 247 for s4 (90–325), and 79, 79 and 80 for s1, s2 and s3, and the variant is the leave-one-variant-out grouping factor. Three tests:

(a) Convergence as label information. A classifier given only each condition's generation and objective reached top-1 0.178 ± 0.013, below uniform chance (0.200) and the majority-class baseline (0.227).

(b) Correlation between convergence and the signal features. Spearman, nine asymmetry features against analysed generation:

feature	**ρ**	**p**
A::ankle_angle_ROMd	−0.304	0.013
A::knee_angle_mind	+0.289	0.019
A::ankle_angle_mind	+0.231	0.062
A::knee_angle_ROMd	−0.191	0.125
A::hip_flexion_mind	+0.152	0.223
A::hip_flexion_ROMd	−0.130	0.298
A::hip_flexion_maxd	−0.125	0.316
A::ankle_angle_maxd	−0.081	0.520
A::knee_angle_maxd	+0.066	0.600

Two of nine reach $p < 0.05$ uncorrected; neither survives Bonferroni ($\alpha = 0.0056$) or Benjamini–Hochberg at 0.05. The associations are weak, but not zero.

(c) The curve with the deeply optimised variants removed, all on the same run:

Capture error	**all five**	**base removed**	**Δ**	**base + s4 removed**	**Δ**
0°	0.837	0.769	−0.068	0.737	−0.100
2°	0.837	0.769	−0.068	0.743	−0.094
5°	0.659	0.692	+0.033	0.599	−0.060
8°	0.591	0.625	+0.034	0.539	−0.052
12°	0.545	0.543	−0.002	0.500	−0.045

Dropping base alone moves the curve at most 0.068, with no consistent sign; dropping base *and* s4 lowers it at every level by 0.045 to 0.100. The two deeply optimised variants were the *easier* ones: depth sets the level of the curve. From 0° to 12° accuracy falls −34.9 % (all five), −29.4 % (base removed) and −32.2 % (both removed), ending at 2.40×, 2.39× and 2.20× the 0.227 majority-class baseline. Only the base-plus-s4 curve is non-monotone, rising 0.006 between 0° and 2°, below the 0.015 interpretability floor, since one of 66 conditions changing its answer moves any accuracy by 1/66 = 0.015, and differences below that are reported but not interpreted. Depth shifts where the curve sits; it does not produce it.

This test uses four classifier seeds against the main pipeline's eight: 0.837 at 0° against the headline 0.845, and 0.659 at 5° in both (S4). s2 is the lowest-strength variant (0.85, table below); removing it entirely moves overall accuracy at 2° from 0.824 to 0.789, a change of −0.035 that exceeds the interpretability floor.

The explanation offered when the dataset grew from 61 to 71 conditions was that the added cells were concentrated in s2. Three predictions follow: s2 is the hardest variant; removing it restores the earlier numbers; kinetics are worth more within s2. None holds. s2 is identified at 0.773 against 0.644 for s1, removing it lowers rather than restores accuracy, and the marginal value of the force plate within s2 is −0.045, neither the highest nor the lowest of the five. The explanation is withdrawn and the cause of the movement is not established.

Between-variant variation. Per-variant identification at 2° capture error, camera tier against camera-plus-plate:

Body	Strength factor	Camera	+ Force plate	Gain
s2	0.85	0.773	0.727	−0.045
s1	0.95	0.644	0.731	+0.087
base	1.00	0.839	0.723	−0.116
s3	1.05	0.866	0.911	+0.045
s4	1.10	0.973	0.991	+0.018

Identification varies from 0.644 to 0.973 between individuals, wider than between any two capture qualities in the main result. The marginal value of kinetics runs from −0.116 to +0.087 with no monotonic relation to variant strength ($\rho = +0.10$, $p = 0.87$, $n = 5$); the five gains average −0.002, the pooled value at 2° in §3.4.

S3. Feature exclusion and tier assignment

Objective-term exclusion. The optimiser's objective penalises peak ground reaction force and constrains walking speed, so features that *are* objective terms encode convergence depth rather than biomechanics; peak vertical GRF correlated with the objective at $\rho \approx 0.6$. Four were excluded a priori as objective terms: `G::vgrf_peak_l`, `G::vgrf_peak_r`, `G::vgrf_peak_asym`, `C::speed`. Two more tracked convergence depth empirically: `C::stance_frac_l` (Spearman $\rho = +0.407$ with analysed generation, $p = 0.001$) and `C::step_time_r` ($\rho = -0.388$, $p = 0.001$).

The empirical threshold. Their contralateral partners fall clearly below both, `C::stance_frac_r` $\rho = -0.176$ ($p = 0.159$), `C::step_time_l` $\rho = -0.115$ ($p = 0.357$), so the one-sided exclusions are data-supported. But `G::vgrf_mean_l` and `G::vgrf_mean_r`, at $\rho = -0.303$ and -0.300, sit just below `C::step_time_r` and were retained. The operative threshold is therefore between $|\rho| = 0.30$ and 0.39 and was never stated as a number. It was fixed before any capture-tier comparison, so not tuned against the outcome; a stricter reader would additionally lose the two mean-force features. Before the exclusion the force plate appeared to add accuracy at laboratory capture quality that it does not have.

Tier boundary. The spatiotemporal features are computed from vertical ground reaction force here, although a camera can observe the same foot-contact events. Marginal value of the force plate under all three defensible assignments:

Capture error	A: timing = camera, plate precision	B: timing = camera, video precision	C: timing = force-plate tier
0°	+0.034	+0.034	+0.038
2°	−0.019	+0.019	−0.064
5°	+0.110	+0.102	+0.019
8°	+0.163	+0.110	+0.110
12°	+0.186	+0.212	+0.193

All three give a positive contribution at 5–12° and an inconsistent sign at 0–2°, which is why the main text reports the low-error result as indistinguishable from zero. Assignment A corresponds closely to the main analysis (+0.032, −0.002, +0.116, +0.159, +0.182). The statement supported is that the sign pattern above 5° is the same under all three. Same 66-condition dataset as the main text.

Degradation of all channels. All channels, force, timing and kinematic alike, scale with capture quality, so no tier is evaluated undegraded.

S4. Dataset rebuilds and run-to-run variation

Run-to-run variation. Several sections re-evaluate the pipeline with four or six classifier seeds where the main analysis uses eight; noise replicates are four throughout. They read about 0.837 at 0° where the main analysis reads 0.845, and differ in the third decimal. Affected: S2 test (c), S7's condition-level column, and S8's white-noise column with its force-plate gain (+0.163 at 8° against +0.159 in §3.4).

Rebuilds. The dataset was rebuilt twice: once to add ten prematurely terminated conditions, and once to repair a provenance defect in which quality control did not re-evaluate a condition when the optimiser subsequently improved (40 of 61 analysed gaits came from an earlier solution than the one recorded). Every headline was therefore measured three times on independently regenerated data. Camera-tier accuracy and the marginal value of the force plate across the three rebuilds are reproduced as Table 5 of the main text.

Accuracy by severity at 2°

Dataset	20 %	40 %	60 %
61 conditions	0.799	0.864	0.875
71, first rebuild	0.650	0.835	0.917
66, repaired	0.610	0.948	0.963

Every reported *direction* held across all three rebuilds: accuracy declines with capture error and stays well above baseline (monotone at the five sampled levels; the finer sweep in S8 rises 0.015 between 0° and 1°, exactly at the interpretability floor, before falling); severe weakness is identified better than mild; the force plate contributes at high capture error, not at low. *Magnitudes* moved substantially, with mild weakness at 2° ranging 0.610–0.799, the precision 66 conditions supports and no more. Only the marginal value of kinetics at 0–2° changed sign, and the main text reports it as indistinguishable from zero.

Resolution. With 66 conditions, one condition changing its answer moves any accuracy by 1/66 = 0.015; smaller differences are not interpreted anywhere in the main text.

S5. Reported joint-angle error of published capture systems

For orientation; this study does not measure what error a given setting produces.

Condition	Reported sagittal joint-angle error	Source
Marker-based, best case	≤ 2° ("acceptable")	McGinley 2009
Theia3D markerless, knee flexion/extension	3.3° RMS difference vs marker-based, the best lower-limb value in that study (hip flexion/extension 11°, ankle 6.7°, knee rotation 13.2°)	Kanko 2021
OpenCap, healthy adults	4.5° MAE	Uhlrich 2023
Smartphone markerless, normal plus imitated crouch / circumduction / equinus (pooled)	5.8° RMSE, 11.3° mean peak	Horsak 2023
Theia3D markerless, cerebral palsy and chronic stroke	hip RMSE within 2.72° on the more-affected side; transverse-plane RMSE consistently high	Steffensen 2023
Single-camera 2D pose (OpenPose, sagittal, healthy adults)	hip 4.0°, knee 5.6°, ankle 7.4° MAE	Steffensen 2023
Stenum 2021		
Self-occluded limb, single-camera 2D	Bland–Altman SD of bias 40 % (ankle), 69 % (knee) and 108 % (hip) greater than the camera-side limb, a relative increase rather than degrees	Wade 2023
Off-axis camera placement	knee RMSE 15.5–29.4° across four off-axis views, with no on-axis arm	Baldinger 2025 (front lunge, not gait)
A real deployed clinical setting	not reported anywhere	,

Camera count buys almost nothing: OpenCap's kinematic estimates "did not substantially improve when using more than two cameras", its authors concluding two are sufficient for the activities tested (Uhlrich 2023). Clothing matters, but little: sport against street clothing gave hip, knee and ankle RMSD averaging 2.6° for Theia3D (Keller 2022), about that system's own between-session variability. Error grows instead with occlusion, off-axis viewing, single-camera 2D estimation and the severity of the pathology.

S6. Quality control and data provenance

Quality control. `qc_gate.py` rejects a condition if the controller never adapted (best solution at generation 0), if the CMA objective exceeds 5.0, if the pelvis falls below 70 % of its steady-state median height, if less than 3 m is travelled, or if the trial is shorter than 8 s.

Repair 1, stale evaluation. Quality control generated the evaluated trajectory only if the file was absent, so 40 of 61 conditions were analysed from an earlier solution than the one whose generation and objective were reported (median objective difference +0.041, maximum +0.197). No acceptance decision changed. The gate now re-evaluates whenever the best solution changes; the whole chain was regenerated.

Repair 2, prematurely terminated optimisations. Thirteen conditions recorded as failing on objective had reached only generations 0–81, against a reference accepted condition at objective 0.765 by generation 15. The scenario files contain no generation cap: these were terminated when processes were paused to free processor time. Re-simulated unmodified, ten passed quality control; five were later lost, leaving 66.

Monitoring fault. A monitoring script restarted twelve optimisations after the dataset had been frozen; the resulting folder moves destroyed the reproducibility of ten conditions, which were re-evaluated before analysis, and five conditions that had passed quality control were lost.

Remaining unfilled cells. Nine of 75: one (s2, knee extensors, 60 %) failed to converge over 321 generations and can be described as attempted; three (base knee flexors 60 %, s2 hip flexors 60 %, s4 knee extensors 60 %) were terminated at generations 55, 39 and 17 and are reported as not adequately attempted, not as infeasible; the remaining five are the conditions lost to the monitoring fault above.

S7. Cluster structure of the kinetics intervals

The 66 conditions come from five variants, and the variant is also the cross-validation fold, so conditions within a variant share a model, a controller and a held-out fold. A pairs-cluster *percentile* bootstrap over five variants is confined to the convex hull of the five variant means, so when they agree in sign it cannot cross zero whatever the true uncertainty; at 5°, 8° and 12° they do agree, so "excludes zero" would carry no information (`cluster_degeneracy.py`). The main text therefore reports a *t* interval on the five per-variant gains (Table 4), which can fail and does at 0° and 2°.

Capture error	**Gain**	**Resampling conditions**	**Resampling variants**	**Excludes zero**
0°	+0.032	[−0.021, +0.087]	[−0.041, +0.082]	no
2°	−0.002	[−0.081, +0.074]	[−0.071, +0.055]	no
5°	+0.116	[+0.002, +0.231]	[+0.058, +0.179]	yes
8°	+0.159	[+0.049, +0.273]	[+0.076, +0.234]	yes
12°	+0.182	[+0.066, +0.295]	[+0.112, +0.252]	yes

The condition-level column is a second Monte-Carlo draw of the same bootstrap, not the interval quoted in the main text (the *t* interval of Table 4).

The variant-level intervals are *narrower* than the condition-level ones at the three highest error levels, which is unusual: the gain is more consistent between variants than between conditions within a variant. With five clusters, the two sets together support a consistent direction and a magnitude not well determined by this design.

Produced by `clustered_ci.py`.

S8. Structured versus white capture error

The primary analysis uses white noise: zero-mean Gaussian, independent per frame, joint and limb. The sweep was repeated with a structured model of 55 % constant per-trial offset, 30 % slow drift across the trial, 15 % per-frame jitter, at the same nominal magnitudes and extended to 20° to cover the off-axis and occluded conditions of S5.

Nominal error	Camera, white	Camera, structured	Difference
0°	0.837	0.837	0.000
1°	0.852	0.822	−0.030
2°	0.837	0.811	−0.026
3°	0.750	0.780	+0.030
4.5°	0.686	0.746	+0.060
6°	0.652	0.701	+0.049
8°	0.591	0.678	+0.087
11°	0.576	0.633	+0.057
15°	0.530	0.602	+0.072
20°	0.477	0.564	+0.087

Above 3° the structured error is *less* damaging than white noise, by 0.05 to 0.09. Range of motion is a maximum minus a minimum, so a constant offset leaves it unchanged and 55 % of the structured budget is such an offset; drift enters only through the part of the sinusoid inside the trial, while white noise inflates both extremes. White noise is therefore the conservative choice for a range-based feature set. Point-in-time features (angle at initial contact, peak angle at a named event) are *not* offset-invariant, so a tolerance figure should be reported with its feature family attached. The force plate's marginal value is correspondingly smaller under structured error (+0.068 at 8°, +0.170 at 20°) than under white noise (+0.163 at 8°, +0.223 at 20°).

Produced by `tolerance.py` and `tolerance.py --white-only`.

S8.1 Feature perturbation induced by a nominal degree value

All eighteen kinematic features are minima, maxima or ranges. An extremum of *n* noisy samples is not an average, it is displaced outward by roughly $\sigma\sqrt{(2 \ln n)}$, so it responds to σ rather than to $\sigma/\sqrt{n}$. Measured over twelve independent error realisations of each condition:

Feature	induced SD at 2°	at 5°	at 12°	induced / nominal
S::knee_angle_max_r	0.88°	2.01°	4.68°	0.41
S::ankle_angle_min_r	1.10°	2.37°	4.64°	0.47
S::knee_angle_ROM_r	1.46°	3.31°	7.55°	0.67
A::knee_angle_ROMd	2.07°	4.60°	10.46°	0.94

A nominal σ induces roughly 0.4–0.9 σ of feature perturbation rather than the $\sigma/\sqrt{n}$ an averaging argument gives, a factor of six to thirteen larger. The degree axis is therefore not directly interchangeable with a reported joint-angle RMSE, and that is stated as a limitation.

Against the variation the classifier has to exploit, the spread of each feature across the 66 conditions with no error added:

Nominal error	median induced / signal	maximum	features where error dominates
2°	0.19	0.46	0 of 57
5°	0.46	1.05	2 of 57
12°	0.97	2.52	28 of 57

At 2° the error is a minor perturbation on every feature, which is why that level behaves as laboratory quality. At 12° it *matches* the between-condition signal on the median feature and exceeds it on 28 of 57, so the tolerance reported at 12° is obtained under a perturbation as large as the signal.

Produced by `noise_magnitude.py`.

S9. Unimpaired baseline against normative gait

Measured on the unimpaired base model, the healthy solution every weakened condition was warm-started from, over the steady-state window:

Measure	Simulated	Normative band	Verdict
Walking speed	1.07 m/s	1.20–1.45	below
Cadence	93.8 steps/min	100–125	below
Stride length	1.37 m	1.20–1.55	within
Hip range of motion	64.2°	40–50	above
Peak swing knee flexion	58.3°	54–64	within
Ankle range of motion	20.5°	22–29	slightly below

Bands: walking speed, Bohannon and Williams Andrews 2011 (meta-analysis, 41 studies, n = 23,111); cadence and stride length, Winiarski et al. 2019 (n = 20, overground); peak swing knee flexion, Patathong et al. 2023 (n = 98; mid-swing 63.6 ± 4.3° and 63.8 ± 4.6°); remaining joint-angle rows, Kwon et al. 2015 (n = 40, young males), which reports joint angles only and is not a source for speed, cadence or stride length. The two knee-flexion sources disagree. Kwon gives 44.3° slow, 54.3° normal, 53.3° fast against Patathong 63.6–63.8°, so the simulated 58.3° falls between them. Speed and cadence bands describe adults aged roughly 20–65; cohorts over 70 walk more slowly than either (Hollman et al. 2011).

Four of six measures fall outside their bands: about 12 % slower and 7 % lower in cadence than a healthy adult, roughly 30 % more hip excursion, slightly less ankle excursion. This is a known property of planar reflex-controlled walkers, where the hip carries limb advancement a person distributes across arms, trunk and frontal plane.

Two consequences. The severity axis (20 / 40 / 60 % force reduction) is not equivalent to a clinical severity scale, since the baseline is not a normal adult gait. And the exaggerated hip motion means hip-flexor weakness may be easier to identify here than in a person, consistent with hip flexors being the best-identified group at laboratory capture quality (100 % at 2°, Table 3).

S10. Features and classifier settings

Tier	Features
Camera (sagittal kinematics + asymmetry + spatiotemporal)	33
Camera + force plate	51
Kinematics only, no timing	27
Force plate alone	18

Before the objective-term exclusion there were 57 features: 18 per-limb sagittal summaries (`S::`), 9 paretic-minus-non-paretic differences (`A::`), 9 spatiotemporal (`C::`) and 21 kinetic (`G::`). Six were removed by the exclusion rule (S3).

Classifier. ExtraTreesClassifier, 100 estimators, `class_weight="balanced"`, all other scikit-learn defaults, no `max_depth`, no `min_samples_leaf`, no hyperparameter tuning of any kind. One binary ensemble per muscle group via `MultiOutputClassifier`, ranked by predicted probability. Median imputation fitted on training folds only. The absence of tuning removes any inner selection loop, and with it any route by which one tier could be tuned harder than another; the cost is that all reported accuracies are lower bounds.

Sample size per fold. 66 conditions × 4 noise realisations, minus the held-out variant. The five variants contain 14, 13, 11, 14 and 14 accepted conditions, so each training fold holds 52 to 55 conditions, 208 to 220 rows, against 33 or 51 features (`fold_sizes.txt`).

S11. Severity: intervals and the interaction test

Whether capture quality costs the severity strata differently is tested on the loss each condition suffers between 2° and 8°, the two levels at which every stratum is measured, bootstrapped over conditions within each stratum (4000 resamples, 8 classifier seeds):

Stratum	n	2°	8°	mean loss
20 % force loss	25	0.610	0.480	−0.130
40 %	24	0.948	0.562	−0.385
60 %	17	0.963	0.831	−0.132

Comparison	difference in loss	95 % CI	excludes zero
40 % vs 20 %	−0.255	[−0.503, −0.010]	yes
40 % vs 60 %	−0.253	[−0.484, −0.007]	yes
60 % vs 20 %	−0.002	[−0.259, +0.250]	no

Both comparisons involving the moderate case exclude zero; severe and mild are indistinguishable. The upper bounds −0.010 and −0.007 sit at or below the study's 0.015 interpretability floor. Resampling variants gives [−0.450, −0.059] and [−0.517, −0.031], also excluding zero, but a five-cluster percentile interval is confined to the convex hull of the five variant means and cannot fail by much. The *t* interval on the five per-variant losses, the estimator adopted for Table 4, includes zero for both:

Comparison	Per-variant losses	Mean	95 % CI (*t*, five variants)
40 % vs 20 %	−0.500, −0.525, −0.100, +0.025, −0.200	−0.260	[−0.563, +0.043]
40 % vs 60 %	−0.125, −0.250, −0.688, +0.131, −0.550	−0.296	[−0.704, +0.112]

The direction holds in four variants of five; the main text therefore reports the moderate case as the largest observed loss rather than a demonstrated interaction (`severity_clustered.py`).

Composition. The strata are unbalanced across muscle groups: 20 % holds five conditions of each mechanism, 60 % only one knee-extensor and two knee-flexor conditions. Reweighted to equal mechanism weight:

Stratum	raw loss	reweighted	shift
20 %	−0.130	−0.130	0.000
40 %	−0.385	−0.390	−0.005
60 %	−0.132	−0.198	−0.065

Moderate-versus-mild is unchanged (−0.255 raw, −0.260 reweighted); moderate-versus-severe falls from −0.253 to −0.193, so roughly a quarter of it is mechanism mix rather than severity. Both keep their direction.

Intervals on the severity table. Bootstrap over conditions within each severity, 4000 resamples, 8 classifier seeds. Non-overlap between rows is descriptive only, and no claim in the main text rests on it.

Capture error	Force loss	top-1	95 % CI	n
2°	60 %	0.963	[0.897, 1.000]	17
2°	40 %	0.948	[0.870, 1.000]	24
2°	20 %	0.610	[0.425, 0.785]	25
8°	60 %	0.831	[0.654, 0.978]	17
8°	40 %	0.562	[0.380, 0.740]	24
8°	20 %	0.480	[0.290, 0.665]	25

These intervals cover the sampling of conditions only, not the across-rebuild variation of S4, which for mild weakness at 2° spans 0.61 to 0.80 and is the larger uncertainty in that row.

Produced by `severity_interaction.py` and `remaining_checks.py`.

S12. Persistence of the confusion structure

The imposed ground truth invites a circularity objection. It conflates two claims: whether a force reduction produces patient-like gait, and whether the classifier reads a graded deviation rather than a fingerprint.

An independent group using the same tool, Waterval et al., reduced plantarflexor strength by 80 % and reproduced measured gait in sixteen patients with bilateral plantarflexor weakness ($R > 0.64$ hip and ankle kinematics and kinetics, $R < 0.55$ knee kinetics, pathological features above 60 % weakness, a threshold that sits above the step observed here, which falls between 20 % and 40 % (Table 2 of the main text), so the two studies agree that identifiability is severity-dependent without agreeing where the step falls) [9]; that 60 % threshold coincides with mild weakness being unidentifiable here at any capture quality.

A fingerprint is present or destroyed, not gracefully degraded; a graded deviation confuses groups whose mechanics resemble each other, so its confusion structure should persist as accuracy falls. Row-normalised confusion, camera tier, leave-one-variant-out:

Imposed	**→ PF**	**→ DF**	**→ KE**	**→ KF**	**→ HF**
2°					
Plantarflexors	0.90	0.00	0.07	0.03	0.00
Dorsiflexors	0.00	0.83	0.00	0.03	0.13
Knee extensors	0.23	0.10	0.60	0.07	0.00
Knee flexors	0.00	0.14	0.00	0.71	0.15
Hip flexors	0.00	0.00	0.00	0.00	1.00
8°					
Plantarflexors	0.89	0.00	0.10	0.01	0.00
Dorsiflexors	0.12	0.50	0.00	0.17	0.21
Knee extensors	0.40	0.03	0.37	0.20	0.00
Knee flexors	0.00	0.22	0.00	0.57	0.21
Hip flexors	0.20	0.07	0.00	0.14	0.58

Six-seed run, not eight: at 8° the knee-extensor diagonal reads 0.37 here against 0.38 in §3.3, and knee extensors called plantarflexors 0.40 against 0.39; quote the main text.

The same pairs dominate at both levels: knee extensors called plantarflexors (0.23 to 0.40), knee flexors called dorsiflexors or hip flexors, dorsiflexors called hip flexors, and the knee-extensor *column* stays almost empty: nothing is ever called a knee extensor. Spearman correlation of off-diagonal cells between error levels against a null permuting one matrix (2000 permutations):

Comparison	**ρ**	**p**	**null ρ, 95th percentile**
2° vs 5°	0.652	0.0019	0.374
2° vs 8°	0.734	0.0002	0.383
2° vs 12°	0.549	0.0122	0.388

All three exceed the null's 95th percentile: the pattern of mistakes is a stable property of the mechanics, not an artefact appearing and disappearing with the noise level.

No deviation reported here has been matched in *magnitude* against a patient's, and the accuracies are therefore upper bounds of unknown looseness. Only patient data can close that.

Produced by `confusion_structure.py`.

S13. Scripts

Script	Purpose
qc_gate.py	quality control; writes qc_accepted.txt and stamps the evaluated parameter file
build_cache.py	parses accepted trajectories into features_det.csv and signals_raw.csv
noise_envelope.py	capture-error sweep
crossover_ci.py	paired bootstrap confidence intervals on the marginal value of kinetics
clinical_breakdown.py	confusion matrix and accuracy by imposed severity
face_validity.py	kinematic signature of each imposed weakness (S1)
face_validity_all.py	whole-cycle face validity, reported for comparison (S1.2)
face_validity_phase.py	face validity measured in the correct gait phase (S1)
exemplar_pf60.py	the plantarflexor exemplar quoted in §3.1
noise_magnitude.py	induced vs signal perturbation per feature (S8.1)
severity_interaction.py	paired test of the severity interaction (S11)
confusion_structure.py	persistence of the confusion pattern (S12)
fold_sizes.txt	per-variant training fold sizes (S10)
s2_live_values.txt	remove-s2 effect and all nine depth correlations (S2)
per_body_2deg.txt	per-variant identification and force-plate gain (S2)
confound_test.py	optimisation-depth confound tests (S2)
why_it_moved.py	tests of the candidate explanations for the s2 shift; none supported, so the shift is reported unexplained (S2)
make_figure0.py	graphical abstract, drawn from the Figure 1 constants
tier_boundary.py	capture-tier boundary sensitivity (S3)
cluster_degeneracy.py	whether a five-cluster percentile interval can fail at all (S7, S11)
clustered_ci.py	variant-level (cluster) bootstrap for the kinetics comparison (S7)
tolerance.py	structured vs white capture-error structure (S8); also run in white-only mode for the S8 comparison, output tolerance_whiteonly.txt
remaining_checks.py	baseline-gait normative check, feature counts, severity intervals (S9, S10, S11)
severity_clustered.py	severity interaction under the variant-level bootstrap and after equalising the mechanism mix (S11)
single_capture.py	accuracy of one capture against the four the pipeline averages (S14.2)
slope_intervals.py	bootstrap intervals on the per-degree loss-rate contrasts (S14.1)
verify_new_refs.py	Crossref check of the sources added to this supplementary (S15)

S14. Additional sensitivity checks

Two further checks: the shape of the decline, and the pooling of noise realisations.

S14.1 Per-degree loss rates

The four per-degree rates are finite differences of five point estimates over unequal spans (2/3/3/4°) whose edges are the convention's own boundaries. `slope_intervals.py` bootstraps the contrast between the 2–5° rate and each neighbour, under both resampling units.

Contrast	Observed	95 % over conditions	95 % over variants
2–5° vs 0–2°	−0.0445	[−0.0868, −0.0013] excludes zero	[−0.0918, +0.0208] includes zero
2–5° vs 5–8°	−0.0354	[−0.0852, +0.0107] includes zero	[−0.0853, +0.0103] includes zero
2–5° vs 8–12°	−0.0421	[−0.0797, −0.0054] excludes zero	[−0.0768, −0.0095] excludes zero

The band is separable from the flat tail but not from the adjoining 5–8° band, and its separation from 0–2° does not survive resampling variants rather than conditions. The ordering is therefore reported as an observation.

S14.2 One capture versus four

Each held-out condition contributes four independent noise realisations and the decision is taken from the mean predicted probability across them (`voi_analysis.loso`), four captures pooled, not the single capture a clinic takes. At 0° no noise is added, so the four replicates are identical and the averaging is degenerate there. `single_capture.py` scores each realisation separately and averages the hits.

Capture error	Four captures pooled (reported)	One capture	Difference
0°	0.845	0.845	+0.000
2°	0.824	0.767	−0.057
5°	0.659	0.643	−0.016
8°	0.600	0.553	−0.047
12°	0.549	0.504	−0.045

No conclusion changes: the one-capture curve stays above twice the 0.227 majority-class baseline at every level, ending at 2.2 times baseline at 12° against 2.4 for the pooled curve, with the steepest band 2–5° on both. The pooled figures are the reported ones because every table, figure and interval is computed on them; the single-capture column is given for comparison.

S15. Sources cited in this supplementary

Sources already numbered in the manuscript are given by their manuscript number. The rest are listed in full: every author-year citation in this document resolves to one entry below. Verified against Crossref by DOI (`verify_new_refs.py`).

Cited as	Full reference
McGinley 2009	manuscript [1]
van der Krogt 2012	manuscript [2]
Uhlrich 2023	manuscript [10]
Horsak 2023	manuscript [11]
Cooper 2012	manuscript [12]
Okada 2024	manuscript [13]
Kesar 2009	manuscript [15]
Bohannon 2011	Bohannon RW, Williams Andrews A. Normal walking speed: a descriptive meta-analysis. *Physiotherapy* 2011;97(3):182–189. doi:10.1016/j.physio.2010.12.004
Winiarski 2019	Winiarski S, Pietraszewska J, Pietraszewski B. Three-dimensional human gait pattern: reference data for young, active women walking with low, preferred, and high speeds. *Biomed Res Int* 2019;2019:9232430. doi:10.1155/2019/9232430
Hollman 2011	Hollman JH, McDade EM, Petersen RC. Normative spatiotemporal gait parameters in older adults. *Gait Posture* 2011;34(1):111–118. doi:10.1016/j.gaitpost.2011.03.024
Kwon 2015	Kwon JW, Son SM, Lee NK. Changes of kinematic parameters of lower extremities with gait speed: a 3D motion analysis study. *J Phys Ther Sci* 2015;27(2):477–479. doi:10.1589/jpts.27.477
Patathong 2023	Patathong T, Klaewkasikum K, Angsnuntsukh C, Woratanarat T, Kijkunasathian C, Sanguantrakul J, Woratanarat P. The knee kinematic patterns and associated factors in healthy Thai adults. *BMC Musculoskelet Disord* 2023;24(1):940. doi:10.1186/s12891-023-07081-7
Sun 2018	Sun D, Fekete G, Mei Q, Gu Y. The effect of walking speed on the foot inter-segment kinematics, ground reaction forces and lower limb joint moments. *PeerJ* 2018;6:e5517. doi:10.7717/peerj.5517
Kanko 2021	Kanko RM, Laende EK, Davis EM, Selbie WS, Deluzio KJ. Concurrent assessment of gait kinematics using marker-based and markerless motion capture. *J Biomech* 2021;127:110665. doi:10.1016/j.jbiomech.2021.110665
Keller 2022	Keller VT, Outerleys JB, Kanko RM, Laende EK, Deluzio KJ. Clothing condition does not affect meaningful clinical interpretation in markerless motion capture. *J Biomech* 2022;141:111182. doi:10.1016/j.jbiomech.2022.111182
Stenum 2021	Stenum J, Rossi C, Roemmich RT. Two-dimensional video-based analysis of human gait using pose estimation. *PLoS Comput Biol* 2021;17(4):e1008935. doi:10.1371/journal.pcbi.1008935
Wade 2023	Wade L, Needham L, Evans M, McGuigan P, Colyer S, Cosker D, Bilzon J. Examination of 2D frontal and sagittal markerless motion capture: implications for markerless applications. *PLoS ONE* 2023;18(11):e0293917.

	doi:10.1371/journal.pone.0293917
Baldinger 2025	Baldinger M, Reimer LM, Senner V. Influence of the camera viewing angle on OpenPose validity in motion analysis. *Sensors* 2025;25(3):799. doi:10.3390/s25030799